

Time-series Time Series Forecasting with Prophet

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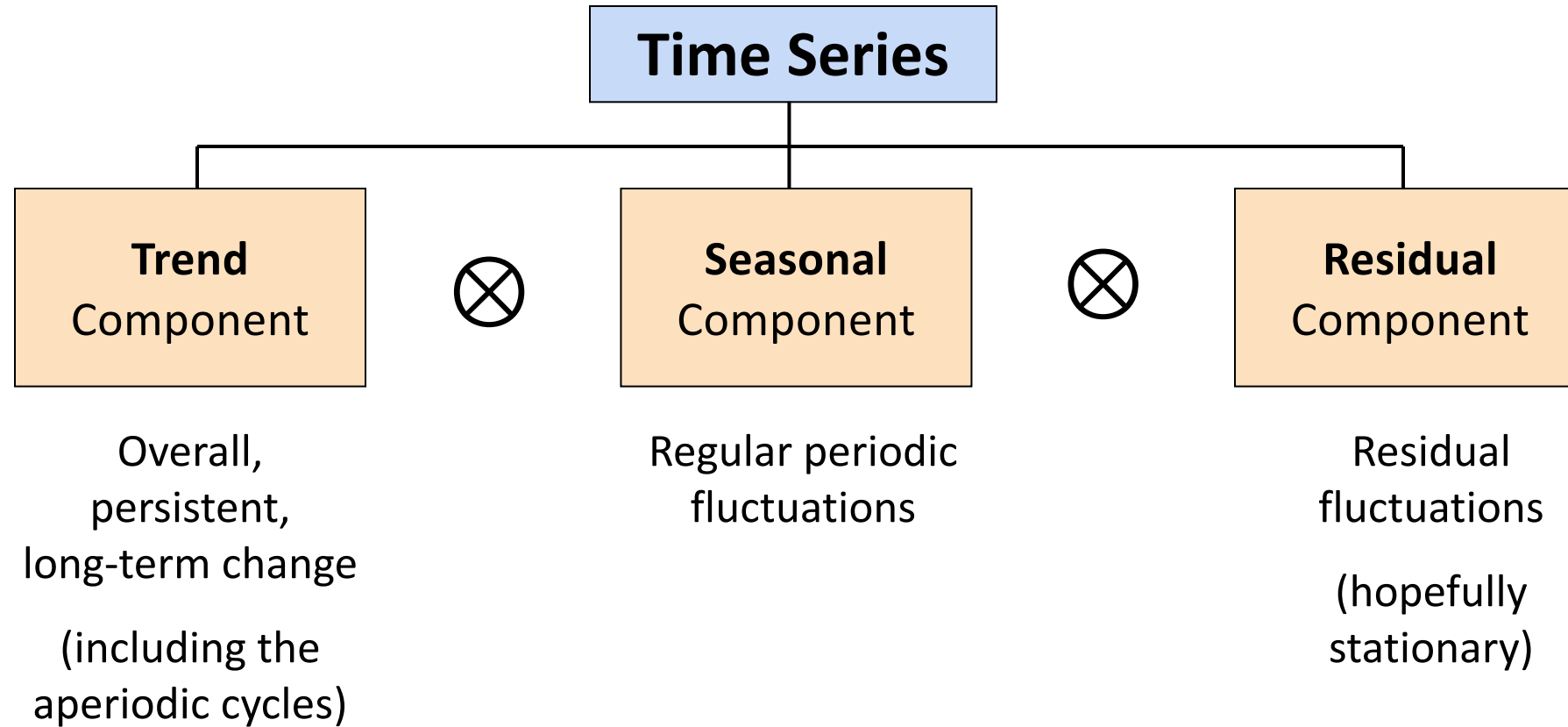
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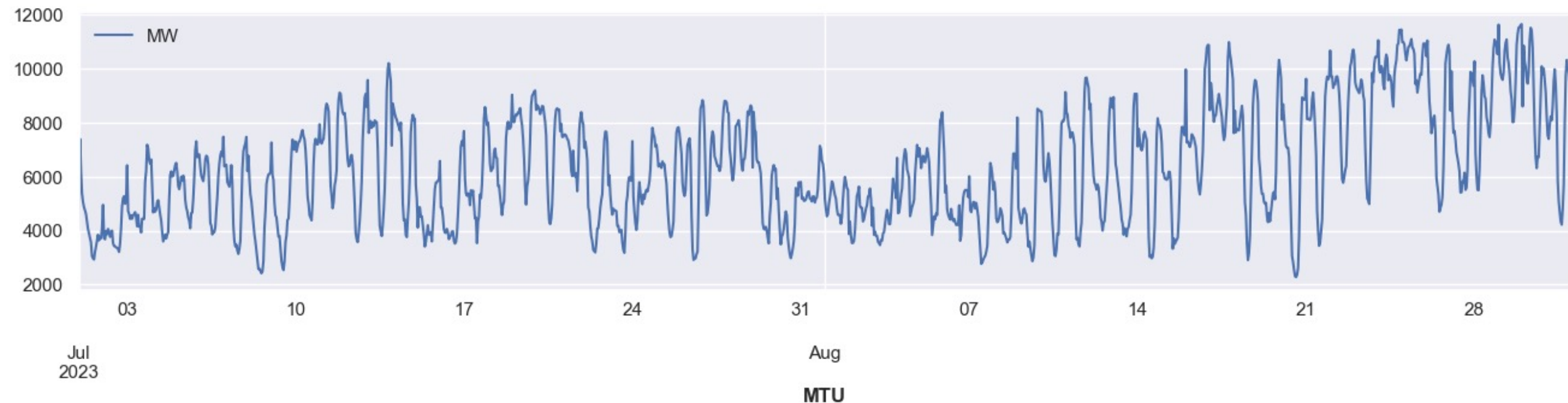
Time-Series Components



 this character is a placeholder for various mathematical operation used to assemble the components

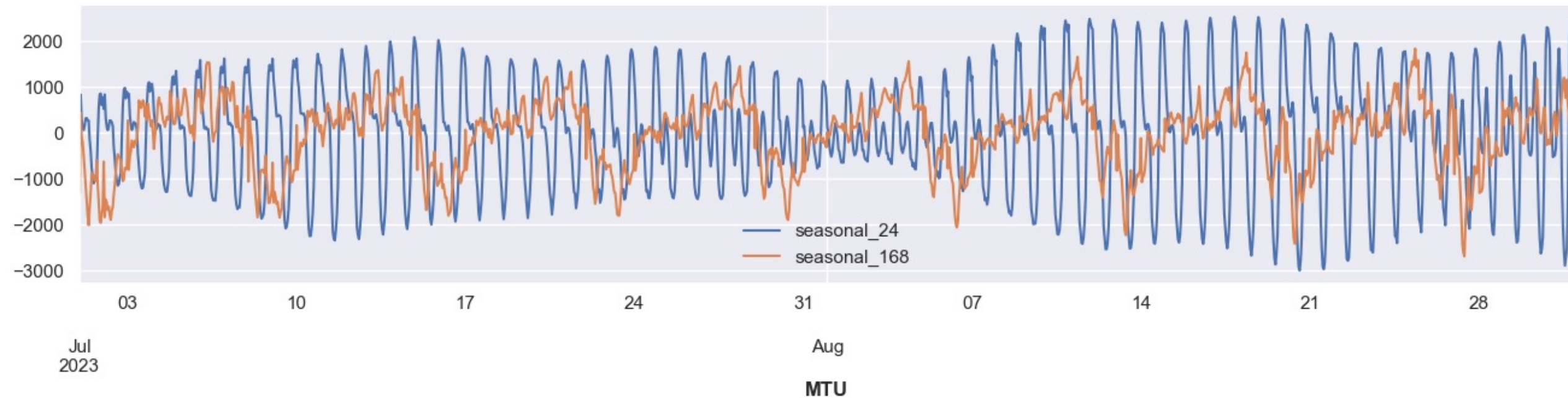
Practical considerations with real-world time series

- 1 trend + 1 seasonality + residual is enough?
- Unfortunately, not all the time series are that simple!



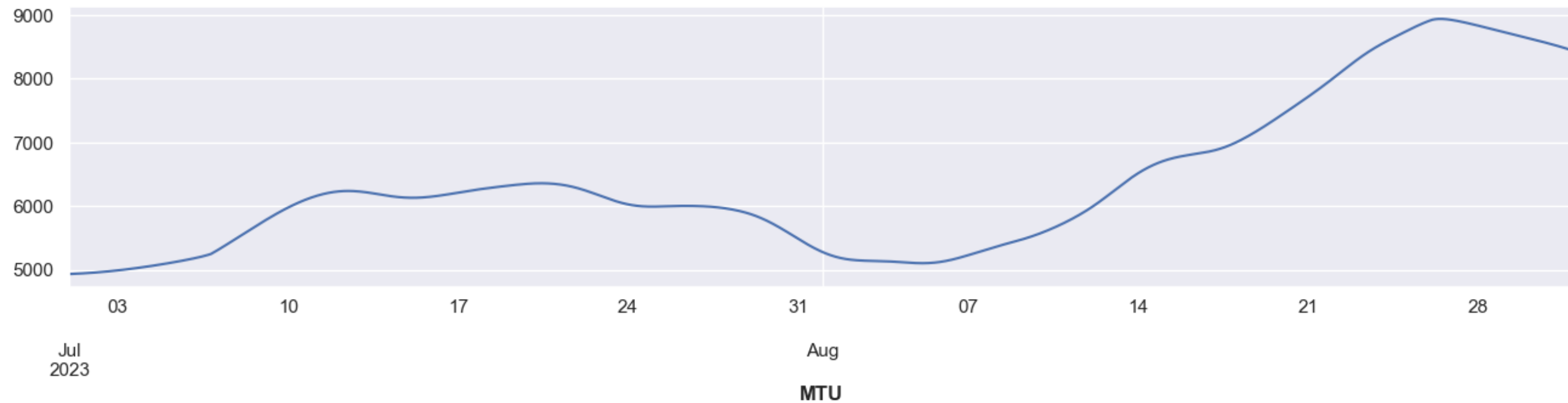
Multiple seasonalities..

- Usually there is more than one seasonality: daily, weekly, monthly, ...



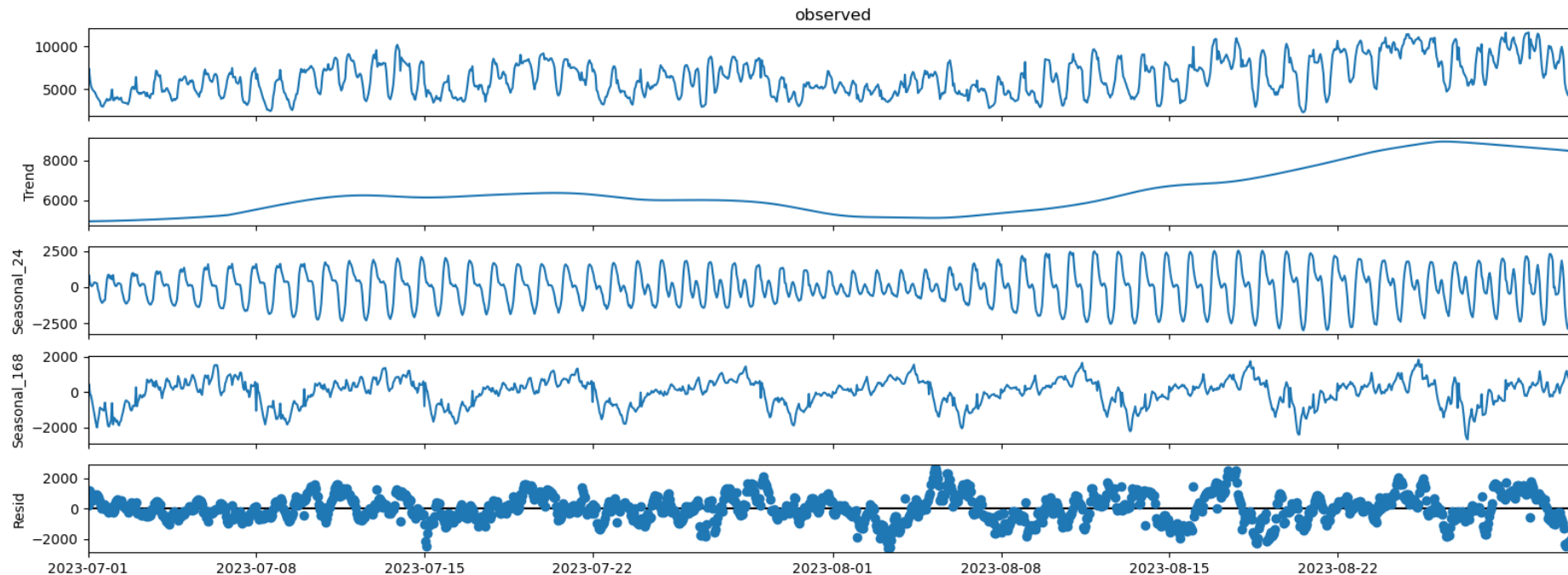
Changing trends..

- Trend may change multiple times throughout the dataset



Advanced decomposition for forecasting

A viable option to model multiple seasonalities is to (i) apply MSTL decomposition (Multiple Seasonal-Trend decomposition), (ii) forecasting each component separately and then (iii) combining forecasts



More advanced statistical forecasting models

There exist some more advanced methods:

- VARIMA - **V**ector **A**uto**R**egressive **I**ntegrated **M**oving **A**verage
- ARCH – **A**uto**R**egressive **C**onditionally **H**eteroscedastic
- GARCH - **G**eneralized **A**uto**R**egressive **C**onditionally **H**eteroscedastic
- BATS - **B**ox-Cox transformation, **A**RMA errors, **T**rend and **S**easonal components
- TBATS - **T**rigonometric seasonality, **B**ox-Cox transformation, **A**RMA errors, **T**rend and **S**easonal components

Still at least some of the characteristics mentioned before must be handled manually by the data scientist..

Many characteristics to take into account!

- Daily seasonality
- Weekly seasonality
- Monthly seasonality
- Yearly seasonality
- Changing trends
- Holiday spikes
- Missing data
- Etc ...



PROPHET

Prophet

- Forecasting tool available in Python and R open sourced by  Meta
- Philosophy behind Prophet:
 - Completely automatic forecasting techniques can be brittle, and they are often too inflexible to incorporate useful assumptions or heuristics.
 - Analysts who can produce high quality forecasts are quite rare because forecasting is a specialized data science skill requiring substantial experience.

Prophet is essentially “framing the forecasting problem as a curve-fitting exercise” rather than looking explicitly at the time based dependence of each observation.

How does it works??



Prophet under the hood

At its core, the Prophet procedure is an **additive regression model** with the following main components:

- *A piecewise linear or logistic growth curve trend.* Prophet automatically detects changes in trends by selecting changepoints from the data.
- *A daily, weekly and yearly seasonal component* modeled using Fourier series.
- A user-provided list of important *holidays*.

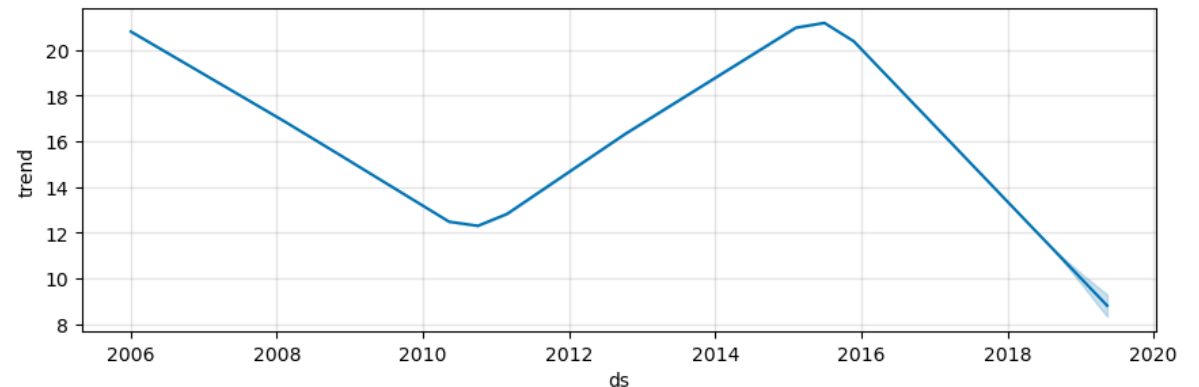
How Prophet works 1/3

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

Noise that cannot be captured by the model

Growth Function (a.k.a. trend)

- Models the non-periodic changes in the time series.
- It captures changing trends through identifying changepoints.
- It is possible to manually set the changepoints.



The Growth Function in detail

- The Growth Function can be a linear or logistic function of time.
- To model non-periodic changes in the trend, the term $a(t)^T$, namely changepoints, was introduced into the equation to allow for changes in the growth factor.
- The logistic function introduces the term $C(t)$ to prevent infinite growth or negative growth in the trend.

Linear

$$g(t) = (k + a(t)^T \delta)t + (m + a(t)^T \gamma)$$

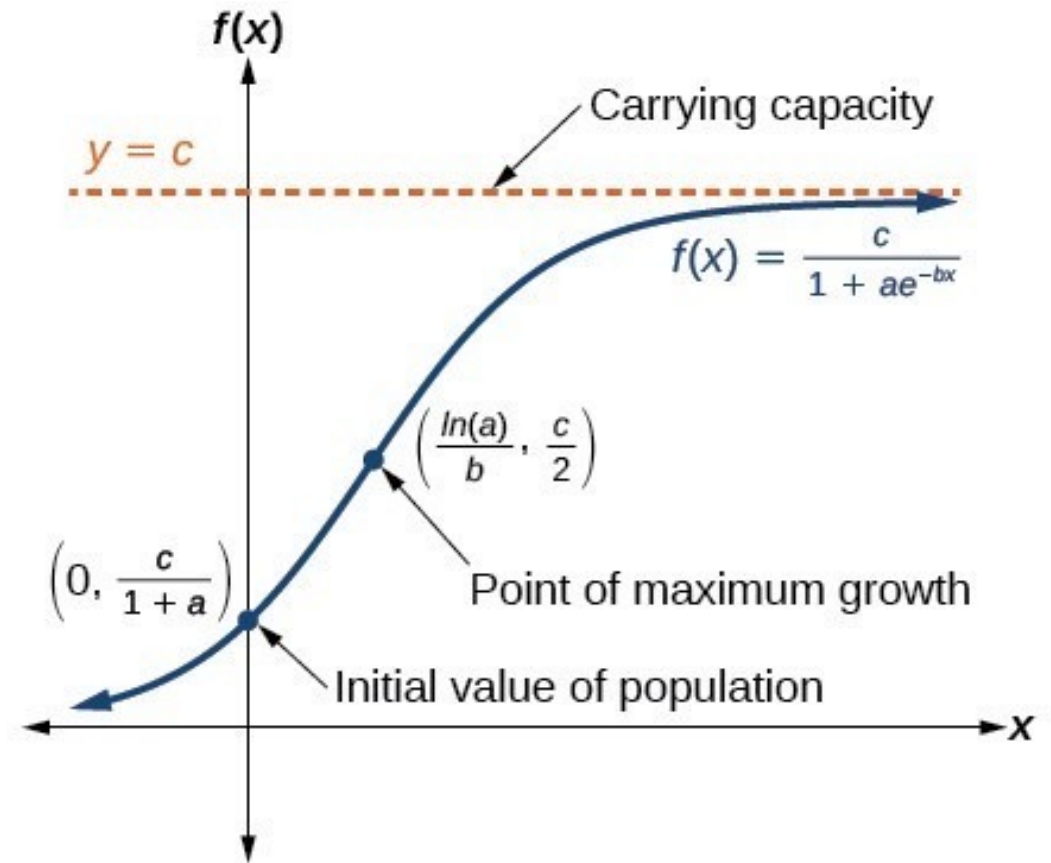
Logistic

$$g(t) = \frac{C(t)}{1 + e^{-(k + a(t)^T \delta)t - (m + a(t)^T \gamma)}}$$

The Growth Function in detail

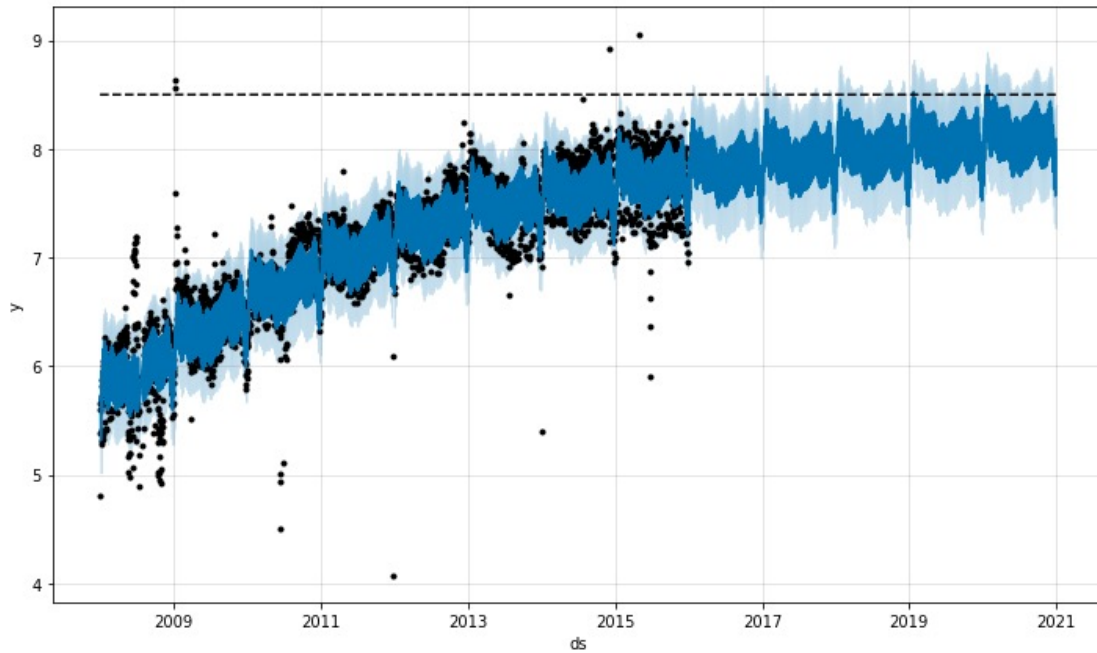
Option 1: Logistic Growth Model

It handles non-linear growth with saturation, i.e., the initial stage of growth is approximately exponential (geometric) as saturation begins, the growth slows to linear (arithmetic) and at maturity, growth stops.

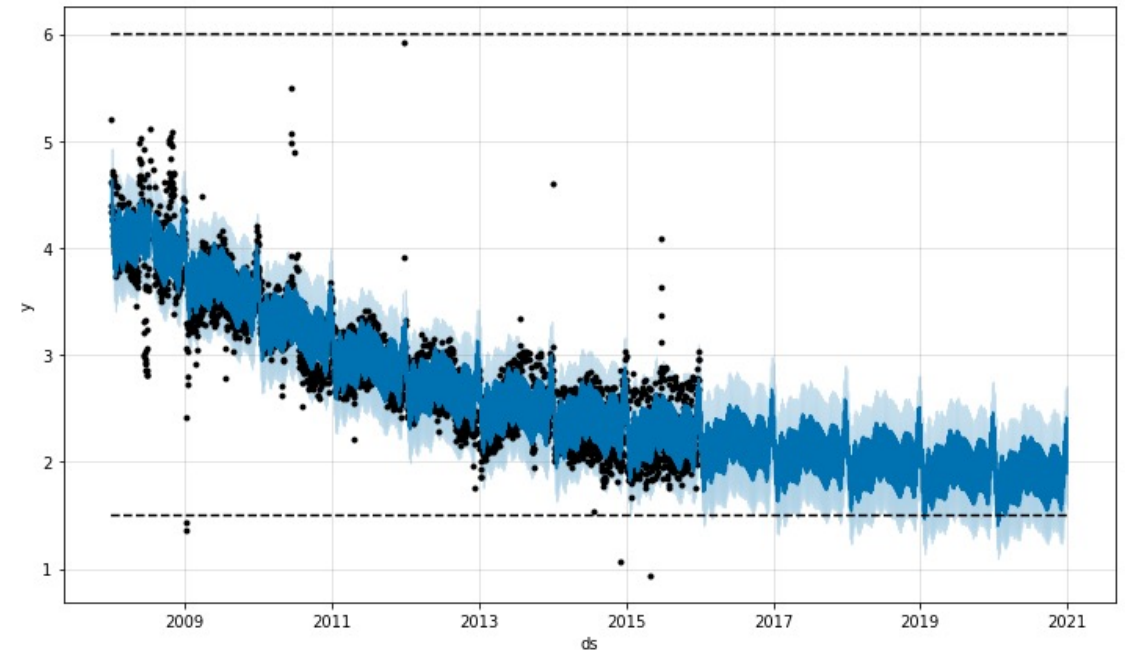


The Growth Function in detail

Forecasting Growth



Saturating Minimum



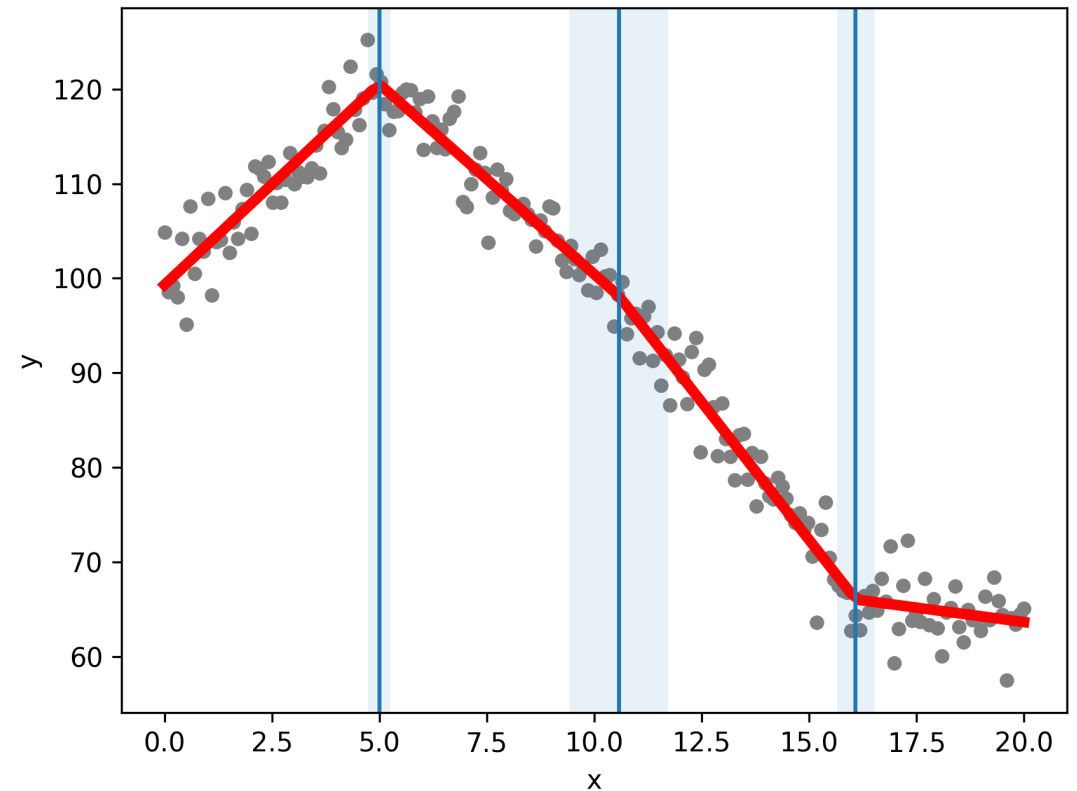
The Growth Function in detail

Option 2: Piecewise Linear Model

It is a simple modification of linear model, yet very useful.

Different ranges of x , different linear relationships occur.

Breakpoints are the value of x where the slope changes



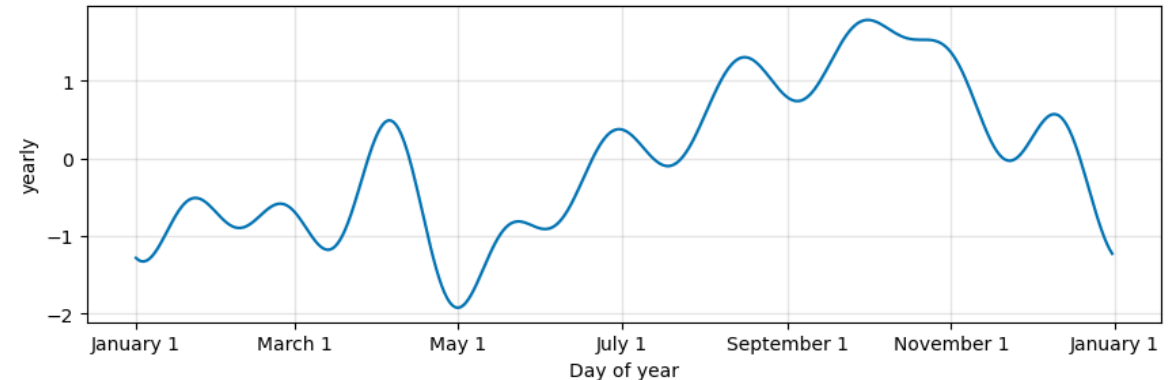
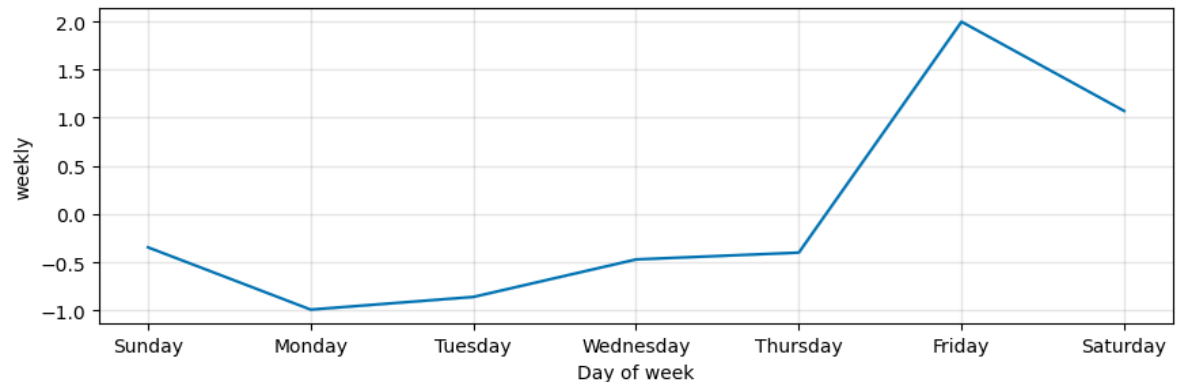
How Prophet works 2/3

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

Noise that cannot be captured by the model

Seasonality Function

- It captures yearly, monthly, weekly, and daily patterns.
- It is estimated through Fourier Series.



The Seasonality Function in detail

Prophet relies on Fourier series to provide a malleable model of periodic effects. The seasonal component $s(t)$ provides a adaptability to the model by allowing periodic changes based on sub-daily, daily, weekly and yearly seasonality.

P is the regular period the time series will have (e.g. $P = 365.25$ for yearly data or $P = 7$ for weekly data, when time is scaled in days).

$$s(t) = \sum_{n=1}^N \left(a_n \cos\left(\frac{2\pi nt}{P}\right) + b_n \sin\left(\frac{2\pi nt}{P}\right) \right)$$

The Seasonality Function in detail

Multiple seasonalities can be modeled with Fourier series with a smoothing prior and a low-pass filter.

Truncating the series at N applies a low-pass filter to the seasonality, so, albeit with increased risk of overfitting, increasing N allows for fitting seasonal patterns that change more quickly.

$$s(t) = X(t)\beta \quad \text{where} \quad \beta = [a_1, b_1, a_2, b_2, a_3, b_3, \dots, a_N, b_N]^T \sim \text{Normal}(0, \sigma^2)$$

For yearly and weekly seasonality $N = 10$ and $N = 3$ respectively can work well for most problems. **The choice of these parameters could be automated using a model selection procedure such as AIC.**

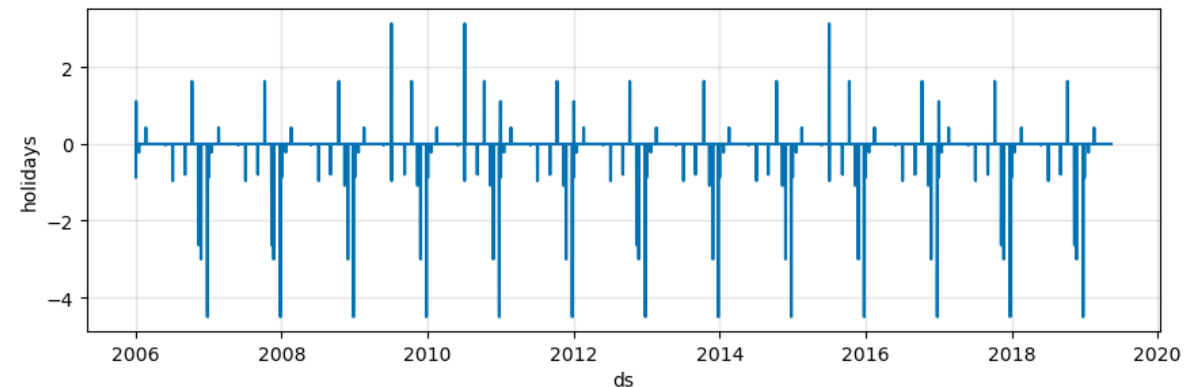
How Prophet works 3/3

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$$

Noise that cannot be captured by the model

Holiday Function

- Add peaks and drops that can be explained by holidays.
- It can account for holidays with dates that changes year to year (e.g., Thanksgiving).



The Holiday Function in detail

Holidays are modeled as independent dummy external regressors allowing the inclusion of a window of days around the holidays to model their effects.

As with seasonality, Prophet uses a prior $\kappa \sim \text{Normal}(0, \sigma^2)$.

$$Z(t) = [1(t \in D_1), 1(t \in D_2), \dots, 1(t \in D_L)] \quad \text{where} \quad h(t) = Z(t)\kappa$$

It is often important to include effects for a window of days around a particular holiday, such as the weekend of Thanksgiving. To account for that we include additional parameters for the days surrounding the holiday, essentially treating each of the days in the window around the holiday as a holiday itself.

Additional regressors

It is also possible to add exogenous variables, i.e., additional regressors.

Prophet provides a more general interface for defining extra linear regressors and does not require that the regressor be a binary indicator. Another time series could be used as a regressor, although its future values would have to be known.

Additional regressors are a generalization of the holidays and events which are binary.

Where Prophet shines

Prophet is optimized for the business forecast tasks at Meta, which typically have any of the following characteristics:

- hourly, daily, or weekly observations with at least a few months (preferably a year) of history
- strong multiple “human-scale” seasonalities: day of week and time of year
- important holidays at irregular intervals known in advance (e.g. the Super Bowl)
- a reasonable number of missing observations or large outliers
- historical trend changes, e.g., product launches or logging changes
- trends that are non-linear growth curves, where a trend hits a natural limit or saturates

Intro to Prophet

1. Predict using Prophet
2. Change weekly and daily seasonality
3. Add holiday effects
4. Analysis of forecast components



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